THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · EASY LEVEL UNIT TEST SET 3

Chapters 5–8: **Describing Motion Around Us, How Forces Affect Motion, Work, Energy and Simple Machines, Sound Waves: Characteristics and Applications** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Scissors are a: 1

(a) first class lever **(b)** pulley
(c) third class lever **(d)** second class lever

2. Echo is due to: 1

(a) reflection of sound **(b)** refraction
(c) dispersion **(d)** absorption

3. Velocity changes from 0 to 10 m/s in 2 s. Acceleration: 1

(a) 20 m/s squared **(b)** 5 m/s squared **(c)** 2 m/s squared **(d)** 10 m/s squared

4. A mass of 10 kg has a weight (g = 10): 1

(a) 1 N **(b)** 10 N **(c)** 100 N **(d)** 1000 N

5. A 2 kg body at 5 m height (g = 10). PE: 1

(a) 10 J **(b)** 20 J **(c)** 100 J **(d)** 50 J

6. Speed = frequency x: 1

(a) wavelength **(b)** mass **(c)** time **(d)** amplitude

7. Velocity = 1

(a) mass x time **(b)** time/displacement
(c) displacement/time **(d)** distance x time

8. Inertia depends on: 1

(a) mass **(b)** volume **(c)** colour **(d)** shape

9. Power = 1

(a) force/time **(b)** work/time **(c)** energy x time **(d)** work x time

10. Sound in air is a: 1

(a) longitudinal wave **(b)** transverse wave
(c) light wave **(d)** radio wave

11. 1 km/h in m/s is: 1

(a) 18/5 m/s **(b)** 10 m/s **(c)** 1 m/s **(d)** 5/18 m/s

12. When a bus starts suddenly we move: 1

(a) forward **(b)** upward **(c)** sideways **(d)** backward

13. A wheelbarrow is a: 1

(a) third class lever **(b)** second class lever
(c) first class lever **(d)** pulley

14. Minimum time gap for an echo is: 1

(a) 10 s **(b)** 0.1 s **(c)** 0.01 s **(d)** 1 s

15. Velocity of 20 m/s for 5 s. Distance: 1

(a) 20 m **(b)** 100 m **(c)** 25 m **(d)** 4 m

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. What is a lever? 2

17. Name two uses of ultrasound. 2

18. Define displacement. 2

19. State Newton's second law. 2

20. Define power. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Give three examples of bodies having PE. 3

22. State three differences between a longitudinal and a transverse wave. 3

23. Convert 20 m/s to km/h. 3

24. Find the momentum of a 5 kg body with a velocity of 4 m/s. 3

OR

Name the three classes of levers with an example each. 3

25. Name three uses of echo. 3

OR

A man walks 4 m east and 3 m north. Find the displacement. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Derive v = u + at. 5

OR

Explain inertia with three examples. 5

27. Derive KE = (1/2) m v squared. 5

OR

Describe the terms of a wave: crest, trough, amplitude, wavelength, frequency. 5

END OF QUESTION PAPER — Total: 50 Marks

NEET TWIN